

9. Which graph type is commonly used in job assignment problems?

A **bipartite graph** is commonly used in job assignment problems because it represents two separate sets:

- **One set:** Jobs/Tasks
- **Other set:** Workers/Employees
- **Edges:** Possible assignments between workers and jobs

20. Which traversal is used to find shortest paths in unweighted graphs?

- **BFS (Breadth-First Search)** is used.
- It works on **unweighted graphs**.
- It uses a **queue**.
- It visits nodes **level by level**.
- It finds the **minimum number of hops**.
- It is useful for **shortest path problems**.
- **Time complexity:** $O(V + E)$.

31. Which algorithm computes all-pairs shortest paths?

Floyd-Warshall algorithm computes the shortest paths between all pairs of vertices in a graph. It uses dynamic programming to find the shortest distance between every pair of nodes. It can handle negative edge weights but not negative cycles. The time complexity is $O(V^3)$.

42. What does a Minimum Spanning Tree minimize?

A Minimum Spanning Tree minimizes the total edge weight while connecting all vertices.

53. In a weighted graph, does the shortest path between two nodes always belong to the MST?

No. The shortest path between two nodes **does not always belong to the MST**. The MST minimizes the **total weight of the tree**, while the shortest path minimizes the **distance between two specific nodes**.

64. If a graph has a negative cycle, what will the diagonal elements of the Floyd -Warshallmatrix eventually show?

The diagonal elements become **negative**.

This indicates a **negative cycle** in the graph.

Thus, the shortest path is **not well-defined**.

75. In a social network graph, what does the "Shortest Path" represent between two users?

In a social network, the **shortest path** represents the **minimum number of connections (hops) between two users**.

86. Which shortest path algorithm is based on the principle of "Successive Approximation"?

Floyd -Warshall algorithm uses the principle of **successive approximation** to find shortest paths.

89. In a bipartite graph, what is the maximum length of an odd cycle?

- A **bipartite graph has no odd cycles.**
- Therefore, the maximum length of an odd cycle is **0**.

92. What is the "In degree" of a vertex in the context of task dependency?

- **Indegree** is the number of incoming edges to a vertex.
- In task dependency, it represents the number of **prerequisite tasks.**
- A task with indegree **0** has no dependencies.